

Chapter 6

Managing Trades

Options traders can hold a position to expiration or close it prior to expiration (active management). Compared to holding a contract until expiration, an active management strategy should be considered for the following reasons:

- It allows for more occurrences over a given time frame (if capital is redeployed).
- It may allow for a more efficient use of portfolio buying power (if capital is redeployed).
- It tends to reduce risk on a trade-by-trade basis.

Trades can be managed in any number of ways, but similar to choosing a contract duration, *consistency* is essential for reaching a large number of occurrences and realizing favorable long-term averages. This book advocates for adopting a simple management strategy that is easily maintainable:

- Closing a trade at a fixed point in the contract duration.
- Closing a trade at a fixed profit or loss target.
- Some combination of these strategies.

This chapter discusses different management strategies, compares trade-by-trade performance, and elaborates on the major factors to consider when choosing appropriate position management. Because management strategies impact the proportion of initial credit traders ultimately collect, the statistics will often be represented as an initial credit percentage rather than dollars. This chapter also predominantly focuses on undefined risk strategy management. Many of these principles also apply to defined risk positions, but defined risk positions are generally more forgiving from the perspective

of trade management because they occupy a smaller percentage of portfolio buying power and have limited loss potential.

Managing According to DTE

As mentioned in [Chapters 3](#) and [5](#), trade profit and loss (P/L) swings tend to become more volatile as options approach expiration. For a strangle, this increase generally results from the price of the underlying drifting toward one of the strikes throughout the contract duration. Consequently, closing a trade prior to expiration, whether at a fixed point in the contract duration or at a specific profit or loss target, tends to reduce ending P/L standard deviation and outlier risk exposure on a trade-by-trade basis. Managing trades actively also frees portfolio buying power from existing positions, which can then be allocated more strategically as opportunities arise. The freed capital can be redeployed to the same type of initial position (increasing the number of occurrences)¹ or to a new position with more favorable short premium conditions (which may be a more efficient use of buying power).

Managing a trade according to days to expiration (DTE), such as closing a position halfway to expiration, offers the benefits described previously and is straightforward to execute. This technique has a clear management timeline and requires minimal portfolio supervision, particularly when portfolio positions have comparable durations. The choice of management time greatly affects the profit potential and outlier risk exposure of a trade because trades managed closer to expiration are more likely to be profitable and have larger profits on average but are generally exposed to more tail risk. The trade-by-trade statistics shown in [Table 6.1](#) compare the performance of different management times for 45 DTE 16  SPY strangles.

Table 6.1 Statistics for 45 DTE 16 Δ SPY strangles from 2005–2021 managed at different times in the contract duration.

16 Δ SPY Strangle Statistics (2005–2021)					
Management DTE	Probability of Profit (POP)	Average P/L	Average Daily P/L	P/L Standard Deviation	Conditional Value at Risk (CVaR) (5%)
40 DTE	67%	2.3%	\$0.23	73%	–206%
30 DTE	73%	10%	\$1.75	88%	–212%
21 DTE	79%	21%	\$1.60	96%	–283%
15 DTE	78%	25%	\$1.51	105%	–304%
5 DTE ^a	82%	33%	\$1.34	185%	–514%
Expiration	81%	28%	\$1.29	247%	–708%

^aStrangles managed at 5 DTE seem to outperform strangles held to expiration because they have a higher POP and average P/L but lower P/L volatility and less tail risk. These results are specific to this strategy and data set, and were likely skewed by significant historical events. This trend is not generalizable across strategies, including the one presented in this table.

Table 6.1 shows that managing a trade prior to expiration is less likely to profit but also has less P/L standard deviation and less tail risk, and it also collects more daily, on average, compared to holding to expiration. These statistics also demonstrate that management time generally carries a trade-off among profit potential, loss potential, and the number of occurrences. Compared to trades managed earlier in the contract duration, trades managed later have larger profits and losses and also allow for fewer occurrences. As early-managed positions accommodate more occurrences and average more P/L per day than positions held to expiration, closing positions prior to expiration and redeploying capital to new positions is generally a more efficient use of capital compared to extracting more extrinsic value from an existing position.

If adopting this strategy, choose a management time that satisfies individual trade-by-trade risk tolerances, offers a suitable profit potential, and occupies buying power for a reasonable amount of time. Remember that selling premium in any capacity carries tail risk exposure even when a position is closed almost immediately (see the 40 DTE results in **Table 6.1**). To achieve

a decent amount of long-term profit and justify the tail loss exposure, consider closing trades around the contract duration midpoint.

Managing According to a Profit or Loss Target

Compared to allowing a trade to expire, managing a position according to a profit target simplifies profit expectations and tends to reduce per-trade P/L variance. Closing limit orders can be set by a trader and automatically executed by the broker, but this management strategy still requires some active maintenance. This is because trades may never reach the predetermined profit benchmark and may require alternative management prior to expiration. Additionally, there is some subtlety in choosing the profit target because that choice significantly impacts the profit and loss potential of a trade, as shown in [Tables 6.2](#) and [6.3](#).

Managing at a profit threshold or expiration generally carries more P/L standard deviation and outlier risk exposure on a trade-by-trade basis than managing at 21 DTE, although it also comes with higher POPs and higher per-trade profit potentials depending on the profit benchmark. Short options are highly likely to reach low profit targets early in the contract duration when P/L swings and tail risk are both fairly low. Therefore, managing a trade according to a low profit target yields a higher strategy POP, lower P/L standard deviation, and less outlier risk compared to managing at a high profit target. However, despite the higher average daily P/Ls, setting the profit threshold *too low* does not allow traders to collect a sufficient credit to justify the inherent tail risk of the position. Average P/Ls are well below the given profit target in all cases due to the tail loss potential. When using a 25% target, for example, the contract failed to reach the target only 4% of the time. Still, those losses were significant enough to bring down the P/L average by more than half. If this management strategy is adopted, a profit threshold between 50% and 75% of the initial credit is suitable to realize a reasonable amount of long-term average profit and reduce the impact of outlier losses. Additionally, because these mid-range profit targets tend to be reached near the contract midpoint or shortly after, these benchmarks also allow for a reasonable number of occurrences.²

Table 6.2 Statistics for 45 DTE 16 Δ SPY strangles from 2005–2021 managed at different profit targets. If the profit target is not reached during the contract duration, the strangle expires. The final row includes statistics for 45 DTE 16 Δ SPY strangles managed around halfway to expiration (21 DTE) as a reference.

16 Δ SPY Strangle Statistics (2005–2021)					
Profit Target	POP	Average P/L	P/L Standard Deviation	Probability of Reaching Target	CVaR (5%)
25% or Exp.	96%	11%	191%	96%	–522%
50% or Exp.	91%	16%	236%	90%	–654%
75% or Exp.	84%	22%	245%	80%	–699%
100% (Exp.)	81%	28%	247%	52%	–708%
21 DTE	79%	21%	96%	N/A	–283%

These tests did not account for whether a P/L target was reached throughout the trading day, but rather whether a target was reached by the end of the trading day. Therefore, these statistics are not entirely representative of this management technique.

Table 6.3 Average daily P/L and average duration for the contracts and management strategies described in [Table 6.2](#).

16 ▲ SPY Strangle Statistics (2005–2021)		
Profit Target	Average Daily P/L Over Average Duration	Average Duration (Days)
25% or Exp.	\$1.75	15
50% or Exp.	\$1.67	24
75% or Exp.	\$1.49	34
100% (Exp.)	\$1.29	44
21 DTE	\$1.60	24

These tests did not account for whether a P/L target was reached throughout the trading day, but rather whether a target was reached by the end of the trading day. Therefore, these statistics are not entirely representative of this management technique. Additionally, because there can be significant variability in when a contract reaches a certain profit threshold, daily P/L estimates were derived from data over the average duration of the trade.

Just as trades can be managed according to a fixed profit target, they can also be managed according to a fixed loss limit (a stop loss). Defining a loss limit is trickier because option P/L swings are highly volatile. Small loss limits are reached commonly, but trades are also likely to recover. Implementing a very small loss limit may significantly limit upside growth and make losses more likely. To understand this, see [Table 6.4](#).

Table 6.4 Statistics for 45 DTE 16 Δ SPY strangles from 2005–2021 managed at different loss limits. If the loss limit is not reached during the contract duration, the strangle expires. The final two rows reference other management strategies for comparison.

16 Δ SPY Strangle Statistics (2005–2021)					
Loss Limits	POP	Avg P/L	P/L Standard Deviation	Prob. of Reaching Target	CVaR (5%)
–50% or Exp.	58%	21%	90%	40%	–168%
–100% or Exp.	69%	25%	110%	25%	–238%
–200% or Exp.	76%	27%	131%	13%	–338%
–300% or Exp.	79%	27%	149%	8%	–450%
–400% or Exp.	79%	27%	160%	6%	–536%
None (Exp.)	81%	28%	247%	N/A	–708%
21 DTE	79%	21%	96%	N/A	–283%
50% Profit or Exp.	91%	16%	236%	90%	–654%

These tests did not account for whether or not a P/L amount was reached throughout the trading day, but rather whether it was reached by the end of the trading day. Therefore, these statistics are not entirely representative of this management technique.

Using a low stop loss threshold, –50% for example, results in lower P/L standard deviation and outlier risk compared to holding the contract to expiration. However, in this case, losses are more common and occur roughly 42% of the time since it is not uncommon for options to reach this loss threshold, although many positions ultimately recover prior to expiration (note the higher POPs for larger limits). Implementing a stop loss also does not necessarily eliminate *all* tail risk exceeding that threshold. For example, despite having a stop loss of –50%, a sudden implied volatility (IV) expansion or underlying price change may cause daily loss to increase

from –25% to –75%, resulting in the closure of the trade but with a final P/L past the loss threshold. Because upside potential is limited and some degree of tail exposure exists with a very small stop loss, a mid-range stop loss of at least –200% is practical.³ Using a stop loss and otherwise holding to expiration generally has a higher profit and larger loss potential than managing at the duration midpoint but tends to carry less tail risk than managing at a reasonable profit target. For more active trading, stop losses are not typically used alone but rather combined with another management strategy.

Comparing Management Techniques and Choosing a Strategy

The strangle management strategies presented thus far are relatively straightforward. These techniques can be ranked according to loss potential (from highest to lowest) and quantified using CVaR and P/L standard deviation of the positions studied:

1. Hold until expiration.
2. Manage at a profit target between 50% and 75%.
3. Manage at a loss limit of –200%.
4. Manage at 21 DTE (halfway to expiration).

Remember that consistency and ease of implementation are important factors to consider when choosing a management strategy. For traders who are comfortable with active trading, strategies can be combined and more precisely tuned according to individual preferences. For instance, suppose a trader of 45 DTE 16 Δ SPY strangles wants a management strategy with a high POP, moderate P/L standard deviation, and moderate outlier exposure. One possibility is managing at 50% of the initial credit *or* at 21 DTE, whichever occurs first. The statistics for this strategy are outlined in [Table 6.5](#).

Table 6.5 Statistics for 45 DTE 16 Δ SPY strangles from 2005–2021 managed either at 50% of the initial credit or 21 DTE, whichever comes first. Statistics for other strategies are given for comparison and ranked by CVaR.

	16 Δ SPY Strangle Statistics (2005–2021)				
Management Strategy	POP	Average P/L	Average Daily P/L	P/L Standard Deviation	CVaR (5%)
21 DTE	79%	21%	\$1.60	96%	–283%
21 DTE or 50% Profit	81%	18%	\$1.67	96%	–288%
–200% Loss or Exp.	76%	27%	N/A	131%	–338%
50% Profit or Exp.	91%	16%	\$1.67	236%	–654%
None (Exp.)	81%	28%	\$1.29	247%	–708%

In this example, the duration and profit targets are moderate, resulting in a combined strategy with smaller but slightly more likely profits than 21 DTE management and significantly less loss potential than 50% profit management. This may be appealing to risk-averse traders because it eliminates a large fraction of the historic losses and significantly reduces tail exposure with the benefit of a slightly higher POP and higher average daily P/L.

When choosing a management strategy, know that all management strategies come with trade-offs among POP, average P/L, P/L standard deviation, and loss potential. How these factors are weighted depends on individual goals:

- For *likely* profits, profit potential must be smaller or exposure to outlier losses must be larger.
- For *large* profits, there must be fewer occurrences or more exposure to outlier losses.

- For a *small* loss potential, profit potential must be smaller or profits must be less likely.

For a qualitative comparison of the different strategies, see [Table 6.6](#).

As mentioned, a suitable management strategy depends on individual preferences for trading engagement, per-trade P/L potential, P/L likelihood and number of occurrences. Following are example scenarios highlighting different management profiles:

Table 6.6 Qualitative comparison of different management strategies.

	Management Strategy			
	21 DTE	50% or Exp.	–200% or Exp.	Exp.
Convenience	Med	High ^a	High	High
POP	Med	High	Med	Med
Per-Trade Loss Potential	Low	High	Low	High
Per-Trade Profit Potential	Med	Low	High	High
Number of Occurrences	Med	Med	Low	Low

^aIf limit orders are used, profit target management is very convenient.

- For passive traders with portfolios that can accommodate more outlier risk, it may make more sense to use only a stop loss and otherwise hold trades to expiration to extract as much extrinsic value from existing positions as possible.
- Active traders with portfolios that can accommodate more outlier risk may manage general positions at a fixed profit target and close higher-risk, higher-reward trades halfway to expiration.
- Very active traders may manage all undefined risk contracts at either 50% of the initial credit *or* halfway through the contract duration because this method prioritizes moderating outlier risk and achieving likely profits of reasonable size.

Generally speaking, an active management approach is more suitable for retail traders because more occurrences can be achieved in a given time

frame, it is a more efficient use of capital, average daily profits are higher, and the per-trade loss potential is lower. It's critical to reiterate that this risk is on a *trade-by-trade* basis. Short premium losses happen infrequently and are often caused by unexpected events, making it difficult to precisely compare long-term performance of strategies of varying timescales. The next section discusses in more detail why comparing the long-term risks for management strategies is not straightforward.

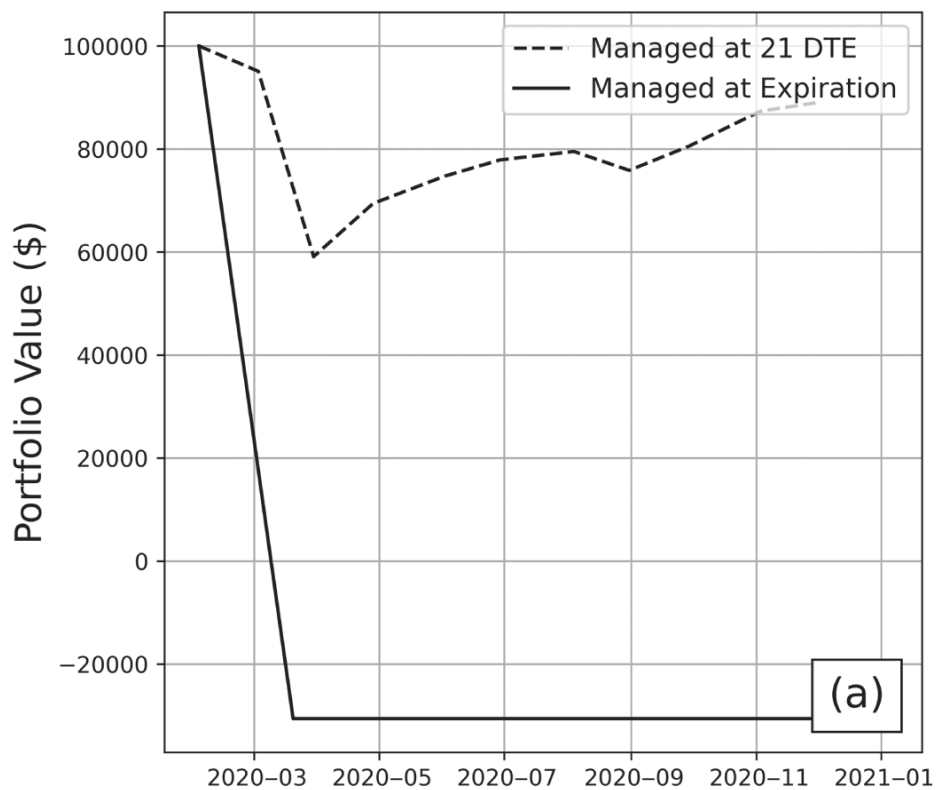
A Note about Long-Term Risk

As mentioned previously, contracts tend to have more volatile P/L swings as the contract approaches expiration. Managing trades prior to expiration, therefore, tends to have lower P/L standard deviation and outlier risk exposure on a trade-by-trade basis compared to holding the contract to expiration. But it's critical to note that this reduction in risk on a trade-by-trade basis *does not necessarily translate to a reduction in risk on a long-term basis*. Though early management techniques reduce loss magnitude *per trade*, inherent risk factors arise from a larger number of occurrences. Consequently, one management strategy may have lower per-trade exposure compared to another, but it may have more *cumulative* long-term risk. Consider the scenarios outlined in [Figures 6.1](#) and [6.2](#). Each scenario compares the performances of two portfolios, each with \$100,000 of capital invested. Both portfolios consist of short 45 DTE 16 Δ SPY strangles continuously traded, but the trades in one portfolio are managed halfway to expiration (21 DTE) and the trades in the other are managed at expiration. The unique market conditions in each scenario affect the performance of each management strategy.⁴

The IV expansion during the 2020 sell-off was one of the largest and most rapid expansions recorded in the past 20 years, producing historic losses for SPY strangles. Due to the timing and duration of this volatility expansion, 45 DTE contracts opened in February and closed at the end of the March expiration cycle experienced the majority of the expansion and were *especially* affected. Shown in [Figure 6.1](#), the portfolio of contracts held to expiration was immediately wiped out by this extreme market volatility, and the portfolio of early-managed contracts incurred a large drawdown of roughly 40% but ultimately survived. This scenario demonstrates how the

loss potential for contracts held to expiration is significantly larger than for contracts managed early. However, this does not necessarily mean that holding to expiration results in more cumulative loss long term.

Management Strategy Comparison (2020)

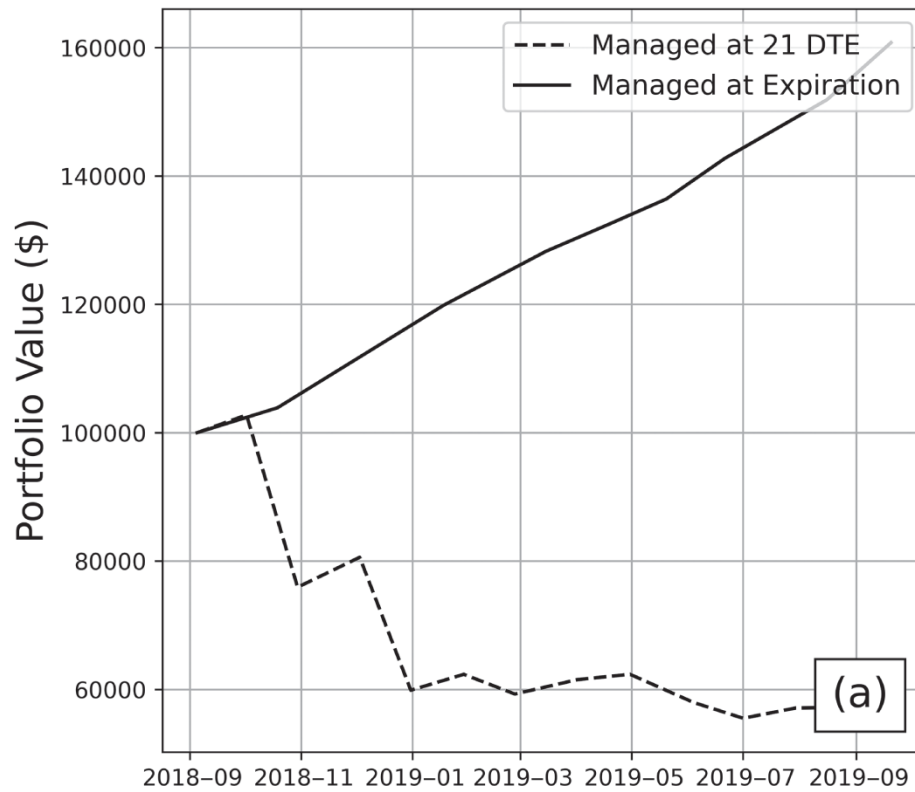


SPY IV



Figure 6.1 (a) Two portfolios, each with \$100,000 in capital invested, trading short 45 DTE 16 Δ SPY strangles from February 2020 to January 2021. One portfolio consists of strangles managed at 21 DTE (dashed line), and the other consists of strangles held until expiration (solid line). (b) The VIX from February 2020 to January 2021.

Management Strategy Comparison (2018–2019)



SPY IV

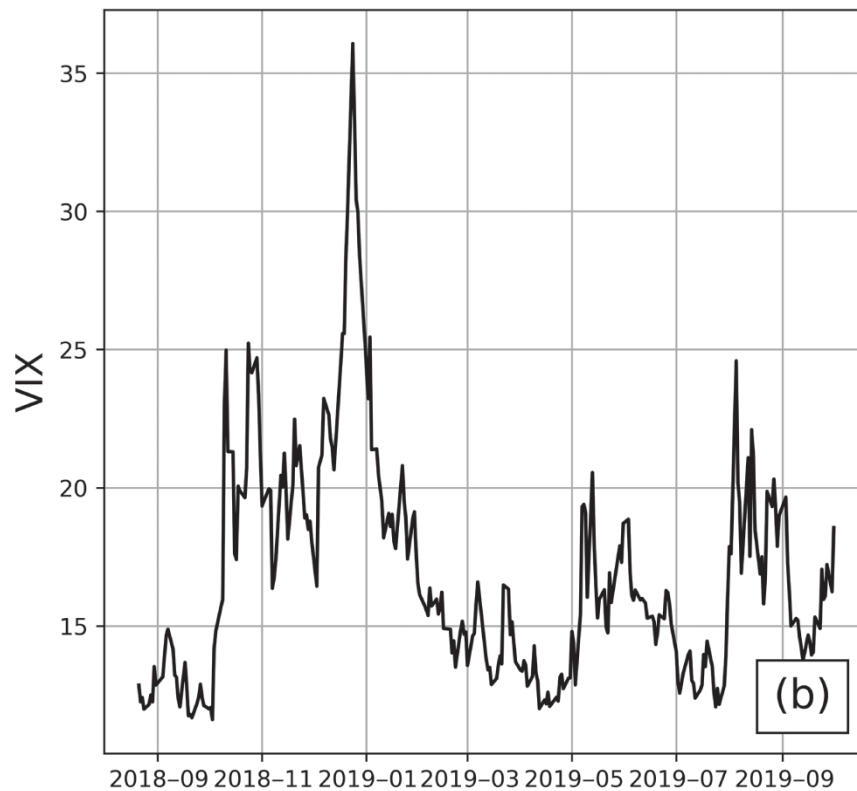


Figure 6.2 (a) Two portfolios, each with \$100,000 in capital invested, trading short 45 DTE 16 Δ SPY strangles from September 2018 to September 2019. One portfolio consists of strangles managed at 21 DTE (dashed line), and the other consists of strangles held until expiration (solid line). (b) The VIX from September 2018 to September 2019.

These same strategies perform quite differently near the end of 2018 when the market experienced smaller, more frequent IV expansions. During this period, the 21 DTE management time for 45 DTE contracts consistently landed on IV peaks during this cycle of market volatility, causing the early-managed portfolio to incur several consecutive losses. Comparatively, the 45 DTE expiration cycles were just long enough to evade these smaller peaks and the portfolio of contracts held to expiration had much stronger performance overall. This scenario demonstrates how having lower per-trade loss potential does not guarantee stronger long-term performance or smaller drawdowns.

Comparing the long-term risks of strategies that occur over different timescales is complicated. These examples show potential trading strategies during unique macroeconomic conditions, but any number of factors could have impacted the realized experience of someone trading during these periods. For instance, if people began trading short 45 DTE 16 Δ SPY strangles on February 3, 2020, they would have had a final P/L of $-\$717$ if they managed at 21 DTE and a final P/L of $-\$8,087$ if they held the contract to expiration. If they instead began trading the same strategy *one month later* on March 4, 2020, they would have had a final P/L of $-\$2,271$ if they managed at 21 DTE and a final P/L of $\$518$ if they held the contract to expiration. Strangle risk and performance, particularly during periods of extreme market volatility, are highly sensitive to changes in timescale and IV. There is as much variation in how people choose contract duration, manage positions, and apply stop losses as there are traders. This makes it difficult to model how people *would realistically trade* in a statistically rigorous way and, consequently, creates complications when evaluating the long-term risk of different management strategies.

Rather than factor in long-term risk when selecting a management strategy, the choice should ultimately be based on the following criteria:

- Convenience/consistency.

- Capital allocation preferences and desired number of occurrences.
- Average P/L and outlier loss exposure *per trade*.

Takeaways

1. Traders should choose a *consistent* management strategy to increase the number of occurrences and the chances of achieving favorable long-term averages. Some management strategies include closing a trade at a fixed point in the contract duration, closing a trade at a fixed profit or loss target, or some combination of the two.
2. Compared to trades managed early in the contract duration, trades managed later have larger profits and losses, higher POPs, and allow for fewer occurrences. Early-managed positions accommodate more occurrences and average more P/L per day than positions held to expiration. Closing positions prior to expiration and redeploying capital to new positions is generally a more efficient use of capital compared to extracting more extrinsic value from an existing position.
3. If managing according to DTE, consider closing trades around the contract duration midpoint to achieve a decent amount of long-term profit and justify the tail loss exposure of short premium.
4. To realize reasonable profits and reduce outlier losses, consider a profit threshold between 50% and 75% of the initial credit. A profit or loss target that is too small (say 25% of initial credit) reduces average P/L and per-trade profit potential, and a profit or loss target that is too large does little to mitigate outlier risk.
5. If implementing a stop loss, a mid-range stop loss threshold of at least –200% is practical because there is limited upside potential and still some degree of tail exposure with a very low stop loss.
6. A suitable management strategy depends on an individual's preferences for trade engagement, per-trade average P/L, per-trade outlier risk exposure, and the number of occurrences. Managing undefined risk contracts at 50% of the initial credit or halfway through the contract duration generally achieves reasonable, consistent profits and moderate outlier risk for those more comfortable with active

trading. This policy of trading small and trading often also allows for more occurrences.

7. Comparing long-term risks of trade management strategies is complicated because unexpected events, such as the 2020 sell-off, affect short premium strategies differently depending on the contract duration. For this reason, compare the risk and rewards of different strategies on a trade-by-trade basis and choose one based on convenience and consistency, capital allocation preferences, tail exposure preferences, and profit goals.
8. The concepts outlined in this chapter are specific to undefined risk positions. These management principles can also be applied to defined risk positions, but defined risk positions are generally more forgiving because they have limited loss potential. It is not as essential to manage defined risk losses because the maximum loss is known, and in some cases, it may be better to allow a defined risk trade more time to recover rather than close the position at a loss.

Notes

- 1 This technique is commonly known as rolling.
- 2 For defined risk positions, a profit target of roughly 50% or lower is more suitable because P/L swings are less volatile and higher profit targets are less likely to be reached.
- 3 Stop losses are not suitable for defined risk strategies. As defined risk strategies have a fixed maximum loss, it is best to allow defined risk losers to expire rather than manage them at a specific loss threshold. This gives the position more opportunity to recover.
- 4 Options portfolio backtests should be taken with a grain of salt. Options are highly sensitive to changes in timescale, meaning that a concurrent portfolio with strangles opened on slightly different days, closed on slightly different days, or with slightly different durations may have performed quite differently than the ones shown here.